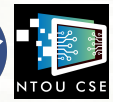
# Auctions & Mechanism Design Basics

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- We study about a kind of science of *rule-making*.
- To make it simple, we first consider single-item auctions.
- We will go over some basics about first-price auctions and second-price auctions.
- Also, we will talk about
  - incentive guarantees,
  - strong performance guarantees, and
  - computational efficiency

  in an auction.
- We will end the discussion with Myerson's Lemma.




# Outline

1. Single-Item Auctions
2. Sealed-Bid Auctions
   - First-Price Auctions
   - Second-Price Auctions
   - Case Study: Sponsored Search Auctions



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# Strategic bidders in an auction

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  - Her maximum willingness-to-pay for it.
  - $v_i$ is private.
    - Unknown to the seller and other bidders.



# What does a bidder want? What's her utility?

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# Sealed-Bid Auctions

## Sealed-Bid Auction

(i) Each bidder $i$ privately communicates a bid $b_i$ to the seller—in a sealed envelope.

(ii) The seller decides who gets the item (if any).

(iii) The seller decides the selling price.



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- Step (ii): The selection rule. We consider giving the item to the **highest** bidder.



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# First-Price auction

**First-Price**

The winning bidder pays her bid.

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- But it's hard to reason about.
- Why?

# Issues of the First-Price Auctions

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# Issues of the First-Price Auctions

- For a bidder: Hard to figure how to bid.
- For the seller: Hard to predict what will happen.



# An Example

- Suppose that you are participating in the first-price auction.
- Your valuation for the item: the number of your birth month + the day of your birth.



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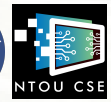
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  - Would it help to know your opponent's birthday?
  - Would your answer change if you knew there were two other bidders rather than one?

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# eBay/Yahoo auction

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    whichever comes first.
  - For example, if the highest other bid is \$90.
    You only pay \$90 $+ \epsilon$ for some small increment $\epsilon$.

  $\approx$ highest other bid!

# Second-Price auction

## Second-Price/Vickrey Auction

The highest bidder wins and pays a price equal to the second-highest bid.

- Is such a strategy a dominant strategy?



# Second-Price auction

**Second-Price/Vickrey Auction**

The highest bidder wins and pays a price equal to the second-highest bid.

- Is such a strategy a dominant strategy?
  - The strategy is guaranteed to **maximize** a bidder's utility **no matter what other bidders do**.



# Truthfully Bidding Is Dominant Here

**Proposition (Incentives in Second-Price Auctions)**

In a second-price auction, every bidder $i$ has a dominant strategy: set the bid $b_i = v_i$, equal to her private valuation.



# Proof of the Proposition

- Fix a bidder $i$ with valuation $v_i$.
- $\boldsymbol{b}$: the vector of all bids.
- $\boldsymbol{b}_{-i}$: the vector of $\boldsymbol{b}$ with $b_i$ removed.

* **Goal**: Show that bidder $i$'s utility is maximized by setting $b_i = v_i$.



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**Proposition 2 (Nonnegative Utility)**

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# Second-Price Single-Item Auctions are "ideal"

**Definition (Dominant-Strategy Incentive Compatible)**

An auction is dominant-strategy incentive compatible (DSIC) if

- truthful bidding is a dominant strategy for every bidder, and
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The social welfare of an outcome of a single-item auction is

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- So such an auction is welfare maximizing if bids are truthful.



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**Theorem**

A second-price single-item auction satisfies:

(1) DSIC.

(2) Welfare maximizing.

(3) It can be implemented in polynomial time.

In fact, (3) is linear.



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# Background

## The Social Dilemma (2020) - Trailer

- Web search results:
  - relevant to your query (by an algorithm, e.g., PageRank).
  - pops out a list of sponsored links.
    - They are paid by advertisers.
- Every time you give a search query into a search engine, an auction is run in real time to decide
  - which advertiser's links are shown,
  - how these links are arranged visually,
  - what the advertisers are charged.




# Multiple Items for Sponsored Search Auctions

- Let's say the items for sale are $k$ "slots" on a search results page.
- Bidders: the advertisers who have a bid on the keyword that was searched on.
  - On the keyword, "university", NTU, NYCU, NCKU, NTOU, etc., might be the bidders.



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  - On the keyword, "camera", Nikon, Canon, Sony, etc., might be the bidders.
  - On the keyword, "SUV", Toyota, Ford, Honda, Porsche, etc., might be the bidders.
- Let's say the items are not identical.
  - Higher slots are more valuable. What do you think?



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- Consider the click-through-rates (CTRs) $\alpha_j$ of slot $j$.
  - The probability that the user clicks on this slot.
  - Assumption: $\alpha_1 \geq \alpha_2 \geq \ldots \alpha_k$.



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  - Here we simply assume that $\beta_i = 1$ for each $i$.
- The expected value derived by advertiser $i$ from slot $j$: $v_i \alpha_j$
- The social welfare is $\sum_{i=1}^{n} v_i x_i$.
  - $x_i$: the CTR of the slot to which bidder $i$ is assigned.
    - $x_i = 0$: bidder $i$ is not assigned to a slot.
  - Each slot can only be assigned to one bidder; each bidder gets (at most) one slot.



# Our Design Approach

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- Who wins what?
- Who pays what?
- The payment.
  - If the payments are not just right, then the strategic bidders will game the system.



# Our Design Approach

## Design Steps

(a): Assume that the bidders bid truthfully. Then, how should we assign bidders to slots so that property (2) and (3) holds?

(b): Given the answer of above, how should we set selling prices so that property (1) holds?



# Step (a)

- Given truthful bids. For $i = 1, 2, \ldots, k$, assign the $i$th highest bid to the $i$th best slot.



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- You can prove that this assignment achieves the maximum social welfare as an exercise.



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- There is an analog of the second-price rule.
  - DSIC.
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    - A powerful and general tool for implementing this second step.
  - Why not just ask the bidders to pay the bid which is next (lower) to its own?
    - ▷ Generalized Second Price Auction.

